Data Analyst/BI Analyst Roadmap

Rare

- **MIS Analyst/MIS Executive**
- **Data Analyst:** Some companies prefer only Excel and SQL skills.

- **MIS Analyst/MIS Executive**
- **Data Analyst/BI Analyst**

(In such instances, it's possible that they maintain a separate DE team or utilize alternative data sources)

- **Data Analyst:** Some companies specifically seek candidates with skills in SQL and Python.

Excel (Required) | SQL (Required) | Power BI/Tableau (Required) | Python (Good to have) | AWS/Azure (Next level)

Frequent

- **Data Analyst/BI Analyst**
- **Power BI Developer**
- **Tableau Developer**

- **MIS Analyst/MIS Executive**

- **Data Analyst/BI Analyst**
- **Data Scientist:** Requires proficiency in statistics, machine learning, and a portfolio of projects to showcase.

- **Data Analyst/BI Analyst**
- **Data Scientist:** Many cloud platforms offer pre-built ML services and APIs, simplifying model development and deployment.
- **Data Engineer:** Proficiency in SQL, Python, and at least one Cloud Platform is the minimum requirement to become a DE.

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Tool 1: Microsoft Excel (Required)

Excel

Why Excel should be the first tool to start as a data analyst:

Foundational Data Skills: Excel equips you with foundational data analysis skills, such as sorting, filtering, lookups, and basic calculations. Learning these concepts lays the groundwork for more advanced analytical techniques.

Data Cleaning and Transformation: Excel's functions and formulas allow you to clean and transform raw data efficiently. You can remove duplicates, handle missing values, and format data for analysis, ensuring your datasets are accurate and usable.

Pivot Tables: Mastering pivot tables in Excel enables you to summarize data, uncover insights, identify patterns, and create meaningful visualizations. This capability provides valuable insights to stakeholders**.**

Managing Large Data, Automating Reporting, and ETL: Power Query in Excel can handle a large volume of data (over 1 million rows). With Power Query, you can automate every step of your data cleaning process and create an automated report that requires just 1 click to refresh, all without needing any code.

Excel can help you secure a job: If you are proficient in Excel and have graduated in any field, you have the potential to land a role as an MIS Analyst solely based on your Excel skills. MIS stands for Management Information System, and this position involves intricate data analysis using Excel as the primary tool.

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SQL

Why SQL should be the 2nd tool after Excel:

Data Management and Scalability: Excel is great for managing and analyzing small to medium-sized datasets, but it can struggle with larger datasets due to performance limitations. SQL databases are designed to handle much larger volumes of data efficiently.

It means you can query a large volume of data and do the exact same thing like sorting, filtering, joining (Lookup functions in Excel), and basic calculations using SQL without sacrificing performance.

Please Note: Keep the advantages of SQL on the side for now. If you're looking to enter the field of Data Analytics, you'll need to pass the machine round of SQL during the interview.

Data Analyst, Data Engineers, and even Data Scientists spend a significant amount of their time working with SQL. SQL is a common tool that is extensively used in these roles.

Some companies only require a Power BI/Tableau Developer with basic SQL skills, but these individuals are extremely proficient with Power BI/Tableau and possess comprehensive knowledge. However, they are also required to successfully pass the SQL test most of the time.

If your goal is to become a **Microsoft Power Platform developer**, then SQL and this roadmap do not pertain to you.

Tool 3: Power BI/Tableau (Required)

Power BI / Tableau

Why Data Visualization tool (Power BI or Tableau):

Data visualization tools are essential for a variety of reasons, such as facilitating data understanding and exploration, insight generation, communicating complex information to both technical and non-technical audiences, storytelling, identification of trends and patterns, decision making, and performance monitoring.

Power BI and Tableau are the two most popular data visualization tools in the job market. I chose Power BI because it required less effort for me to learn due to my proficiency in Power Query and DAX skills from Excel.

Please Note: Mastering SQL and possessing intermediate knowledge of Power BI/Tableau, along with Excel proficiency, are sufficient to secure a job as a Data Analyst or BI Analyst.

However, advanced expertise in any specific data visualization tool offers distinct advantages for effectively presenting your findings. For instance, having proficiency in Advanced DAX empowers you to achieve advanced formatting and visualization within Power BI.

At this stage, you should apply your skills to construct a **project portfolio**. Simultaneously, ensure that your **resume** is prepared (**compatible with ATS**) and consistently **update your profile on Naukri each day**.

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Tool 4: Python (Good to have)

Python

Why Python:

Python is a valuable skill to have, but it's not necessarily a tool you must learn in the first place. Many problems can be solved using SQL and Power BI/Tableau. However, there are numerous challenges that require the use of Python.

If the job posting of a company you are targeting requires Python, then you should consider learning it. While basic Python skills are sufficient for many entry-level positions, having proficiency in Python is often considered a beneficial skill rather than a strict requirement.

Please Note: For a Data Analyst, basic Python along with Pandas, NumPy, and Matplotlib is sufficient.

However, if you intend to transition into Data Engineering or Data Science, then learning Python is necessary. You will also need to search for the most relevant libraries for the specific role of a Data Engineer or Data Scientist.

A Data Analyst with machine learning capabilities and strong statistical knowledge can distinguish themselves from other applicants. However, achieving this level of proficiency takes time. It's important not to overlook the prerequisite skills of Excel, SQL, and a Data Visualization tool.

This is why it's recommended to secure a job first. Without day-to-day application, it can be challenging to retain and enhance your existing skills.

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Tool 5: AWS/Azure (Next level)

AWS / Azure

Why Cloud Platform:

Learning cloud platforms like AWS or Azure can be extremely beneficial for Data Analysts or BI Analysts. Below, you can see how leveraging your existing SQL and Python skills with the cloud can boost your performance and elevate you to the next level

Leveraging SQL on a cloud platform:

- Cloud DBs/services for hassle-free DB management and scaling.
- SQL for intricate queries, joins, insights from big data.
- Analyze data lakes via SQL, great for unstructured data.
- AWS Glue, GCP Dataflow, Azure Data Factory for ETL.
- Blend SQL with serverless (AWS Lambda, GCP/Azure Functions) for auto pipelines, event triggers, on-demand computations.

Leveraging Python on a cloud platform:

- Python + cloud = scale, efficient analysis.
- Libraries (SQLAlchemy, PySpark) for ETL.
- Cloud trains ML models at scale.
- Real-time processing with AWS Kinesis, Azure Stream Analytics, GCP Dataflow.
- Python automates tasks, triggered by cloud events.

Please note: Using these platforms requires a subscription, and many organizations today are utilizing them.

In terms of when to learn these platforms, it's recommended to consider doing so after securing a job and after learning at least SQL.

Ratnesh